Zikai Xiao

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Research Interests

My research primarily focuses on **Large Language Models (LLMs)**, with specific interests in:

- **LLM Agents & Environments**: Web Agent RL, World Models, Generative Environments, File-in-file-out Agent.
- **Long-Context LLMs**: Long Input Understanding, Long-form generation.

Education

- **Zhejiang University** — *Sep 2022 – Jun 2027*
  Ph.D. in Computer Science and Technology — Hangzhou, China
- **China University of Geosciences** — *Sep 2018 – Jun 2022*
  B.Eng. in Automation — Wuhan, China

Experience

- **Qwen Team** — *Dec 2024 – Present*
  Research Intern — Hangzhou, China
  - Core contributor to **Qwen3.7** ("The Agent Frontier" flagship), building the browser-domain agentic RL environment, training data, and LLM-based reward pipeline powering its long-horizon tool-use capabilities.
  - Core contributor to **Qwen-AgentWorld**, the first language world model simulating **7 agentic domains** from 10M+ trajectories; serves as both a controllable RL simulator and an agent foundation model.
  - Built **WebWorld** (ICML 2026), the first open-web world model series (8B/14B/32B, Apache 2.0); synthesized trajectories yield **+9.9% MiniWob++ / +10.9% WebArena**.
  - Led **LongWeave** (EMNLP 2025), a verifiable evaluation benchmark for real-world long-form generation.

Research

- **Agentic RL for Qwen3.7: Playwright MCP Tool-Use** — *Feb 2026 – May 2026*
  Research Intern, Qwen Team Blog — Qwen Blog · Core Contributor
  - Extended **MCPMark**'s **Playwright** environment with Dockerized WebArena-style tasks; built automated data-collection and evaluation agents.
  - Designed **LLM-based Agent Judge** replacing rule-based rewards, correctly rewarding **diverse correct solutions** that rigid rule-based checks would miss.
- **WebWorld: A Large-Scale World Model for Web Agent Training** — *Dec 2024 – Feb 2026*
  Research Intern, Qwen Team Code 10k — ICML 2026 · First Author
  - First open-web world model series (8B/14B/32B, Apache 2.0) with **1M+** real-world web trajectories.
  - Proposed **WebWorld-Bench** with dual evaluation metrics; 32B matches Claude-Opus-4.1 and Gemini-3-Pro.
  - Training yields **+9.9% MiniWob++ / +10.9% WebArena**; **outperforms GPT-5** as lookahead world model.
- **Qwen-AgentWorld: Language World Models for General Agents** — *Oct 2025 – Jun 2026*
  Research Intern, Qwen Team Code · 630★ 18.9k Blog — Qwen Blog · Core Contributor
  - First language world model (**35B-A3B / 397B-A17B** MoE) simulating **7 agentic domains** via long CoT; trained on **10M+** trajectories through CPT→SFT→RL.
  - Proposed **AgentWorldBench** across 9 agent suites; surpasses all frontier models including GPT-5.4.
  - As RL simulator, gains **surpass real-env training**; world-model warm-up also lifts downstream agent benchmarks.
- **LongWeave: A Long-Form Generation Benchmark Bridging Relevance and Verifiability** — *Jan 2025 – May 2025*
  Research Intern, Qwen Team Code — EMNLP 2025 · First Author
  - Proposed **Target-Anchored Synthesis** to bridge verifiability and real-world alignment in long-form generation.
  - Benchmark covers 5 scenarios and 7 task types (0–128k input, 0–8k output), containing 5600 samples.
- **Positional Contrastive Decoding for Enhancing Long-Range Perception in LLMs** — *May 2024 – Mar 2025*
  Zhejiang University Code — ACL 2025 · First Author
  - Identified **Posterior Salience Attenuation (PSA)** causing long-text performance degradation.
  - Proposed **PCD**, a training-free method contrasting long and local attention to mitigate rank collapse.
- **Personalized Federated Learning for Long-Tailed Data (FedLoGe)** — *Jun 2023 – Apr 2024*
  Zhejiang University – Angelalign Inc. R&D Center Code — ICLR 2024 · First Author
  - Proposed **FedLoGe** for improving local and global models under long-tailed federated data.
  - Designed **SSE-C** to prune noisy features and enhance representations.
  - Developed **GLA-FR** for feature realignment, achieving SOTA on CIFAR-10/100-LT and ImageNet-LT.
- **Federated Long-Tailed Learning with Self-Adjusting Gradient Balancer (FedGraB)** — *Sep 2022 – Jun 2023*
  Zhejiang University – Angelalign Inc. R&D Center Code — NeurIPS 2023 · First Author
  - Proposed **FedGraB**, combining Self-adjusting Gradient Balancer (SGB) and Direct Prior Analyzer (DPA) for federated long-tailed learning.
  - Re-weighted client gradients based on global distribution feedback, mitigating data heterogeneity and imbalance.

PUBLICATIONS

C=CONFERENCE, R=REPORT

[C.1] **Zikai Xiao**, Jianhong Tu, Chuhang Zou, Yuxin Zuo, Zhi Li, Peng Wang, Bowen Yu, Fei Huang, Junyang Lin, Zuozhu Liu. (2026). **WebWorld: A Large-Scale World Model for Web Agent Training**. In *Proceedings of ICML 2026*.

[C.2] **Zikai Xiao**, Fei Huang, Jianhong Tu, Jianhui Wei, Wen Ma, Yuxuan Zhou, Jian Wu, Bowen Yu, Zuozhu Liu, Junyang Lin. (2025). **LongWeave: A Long-Form Generation Benchmark Bridging Real-World Relevance and Verifiability**. *EMNLP Findings 2025*.

[C.3] **Zikai Xiao**, Ziyang Wang, Wen Ma, Yan Zhang, Wei Shen, Wang Yan, Luqi Gong, Zuozhu Liu. (2025). **Mitigating Posterior Salience Attenuation in Long-Context LLMs with Positional Contrastive Decoding**. In *Proceedings of ACL 2025*.

[C.4] **Zikai Xiao**, Zihan Chen, Liyinglan Liu, Yang Feng, Jian Wu, Wanlu Liu, Joey Tianyi Zhou, Howard Hao Yang, Zuozhu Liu. (2024). **FedLoGe: Joint Local and Generic Federated Learning under Long-tailed Data**. In *Proceedings of ICLR 2024*.

[C.5] **Zikai Xiao**, Zihan Chen, Songshang Liu, Hualiang Wang, Yang Feng, Jin Hao, Joey Tianyi Zhou, Jian Wu, Howard Hao Yang, Zuozhu Liu. (2023). **Fed-GraB: Federated Long-tailed Learning with Self-Adjusting Gradient Balancer**. In *Proceedings of NeurIPS 2023*.

[R.1] Yuxin Zuo, **Zikai Xiao**, Li Sheng, Fei Huang, Jianhong Tu, ..., Junyang Lin. (2026). **Qwen-AgentWorld: Language World Models for General Agents**. *arXiv Preprint 2026*.

[C.6] Xingyue Huang*, Rishabh*, Gregor Franke*, Ziyi Yang*, **Zikai Xiao**, ..., Guohao Li. (2025). **Loong: Synthesize Long Chain-of-Thoughts at Scale through Verifiers**. *NeurIPS Workshop 2025*.

[C.7] W. Shen, C. Yin, Y. Liu, **Zikai Xiao**, X. He, Y. Wang. (2025). **Why RoPE Struggles to Maintain Long-Term Decay in Long Sequences?**. *ICLR 2025 Blog Track*.